

Lecture 6 — Fourier methods for option pricing

Study notes: the Fourier family, density recovery by COS, and the COS pricing formula

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Abstract

These notes work through Lecture 6 in full. The destination is the *COS pricing formula* for a European option,

$$P(x_0, T) \approx e^{-rT} \sum_{k=0}^{N-1'} \operatorname{Re} \left\{ \varphi_{X_T} \left(\frac{k\pi}{b-a}; x_0 \right) e^{-ik\pi \frac{a}{b-a}} \right\} V_k,$$

and everything needed to reconstruct it on paper: the characteristic function as the Fourier dual of the density, why a *cosine* (not full complex) series, the closed form of the cosine coefficients A_k in terms of the characteristic function φ , the closed form of the payoff coefficients V_k for a call/put, the cumulant-based truncation range $[a, b]$, and the error analysis. The closed forms of V_k and of $[a, b]$ are not on the slides; they are taken from Fang & Oosterlee (2008/2009) and derived here because they are the parts an examiner will press on. The prime on the sum means the $k = 0$ term is halved.

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1 Where COS sits, and what problem it solves

The risk-neutral price of a European-style claim with maturity T , payoff $V(x_T, T)$ and discount rate r is the discounted expectation

$$P(x_0, T) = e^{-rT} \mathbb{E}[V(x_T, T) \mid x_0] = e^{-rT} \int_{\mathbb{R}} V(y, T) f(y \mid x_0) dy, \quad (1)$$

where x_T is the state variable (typically $x_T = \ln S_T$ or the log-return $\ln(S_T/S_0)$) and $f(\cdot \mid x_0)$ is its risk-neutral conditional density. The numerical problem is that f is rarely known in closed form. What *is* known, for an enormous class of models (Black–Scholes, Heston, Variance-Gamma, the affine jump-diffusions, the Lévy models tabulated in §2.2), is the *characteristic function* φ_{X_T} . COS is a way to compute the integral (1) using only φ , with no numerical integration and with (for smooth densities) exponential convergence in the number of terms. It is the method this course is built around.

The logic of the lecture is three-fold and these notes follow it:

1. **The Fourier family** (§2): the continuous Fourier transform, the characteristic function as a special case, the inversion formula, the DFT/FFT, and Fourier(-cosine) series. This fixes the vocabulary and the one identity COS exploits.
2. **Density recovery by COS** (§4): approximate a density on a finite range $[a, b]$ by a cosine series whose coefficients are *sampled directly from* φ .
3. **COS for option pricing** (§5): insert the cosine expansion into (1), swap sum and integral, and read off the pricing formula; then compute V_k in closed form and fix the range $[a, b]$ from cumulants.

2 The Fourier family

2.1 Continuous Fourier transform (CFT)

For integrable $f : \mathbb{R} \rightarrow \mathbb{C}$ one convention of the transform / inverse pair is

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} f(x) e^{i2\pi\omega x} dx, \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{-i2\pi\omega x} d\omega. \quad (2)$$

$\hat{f}(\omega)$ is the amplitude/phase of the frequency- ω component; the inversion rebuilds f as a superposition of complex exponentials of all frequencies. (Conventions differ in the placement of 2π and the sign in the exponent; only internal consistency matters.)

2.2 The characteristic function as Fourier dual of the density

The characteristic function of a random variable X is

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} f_X(x) dx. \quad (3)$$

Comparing (3) with (2), φ_X is (up to the 2π -scaling convention) the Fourier transform of the density f_X . The CDF F_X determines φ_X and vice versa; when a density exists, f_X and φ_X are a Fourier pair. The inversion reads

$$f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \varphi_X(t) dt. \quad (4)$$

This is the integral COS evaluates cheaply. Before COS, (4) (or the equivalent pricing integral) was attacked by direct quadrature or by FFT (Carr–Madan, §3).

Remark 1 (Why the characteristic function is the right object). *The whole appeal is in §2.2: a long list of pricing models give φ in closed (or semi-closed) form even when f is intractable — Black–Scholes, Heston, Variance-Gamma, Merton/Kou jumps, the affine jump-diffusions, CGMY, NIG. Some of these are tabulated in §9. A method that needs only φ inherits this entire library for free.*

2.3 Discrete Fourier transform (DFT) and the FFT

On a sequence $\{x_n\}_{n=0}^{N-1}$ the DFT and its inverse are

$$X_k = \sum_{n=0}^{N-1} x_n e^{i\frac{2\pi}{N}kn}, \quad x_n = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{-i\frac{2\pi}{N}kn}. \quad (5)$$

Computed naively this is $O(N^2)$. The *Fast Fourier Transform* factorises the DFT matrix into $O(\log N)$ sparse factors (the radix-2 split into even/odd indexed sub-DFTs of length $N/2$), giving $O(N \log N)$. The FFT is what makes the Carr–Madan approach practical and is the benchmark COS is measured against. Note for later: COS does *not* use the FFT — its cost is the bare $O(N)$ evaluation of the single sum.

2.4 Fourier series and the half-range cosine series

A 2π -periodic function expands as $f(x) = \frac{a_0}{2} + \sum_{n \geq 1} (a_n \cos nx + b_n \sin nx)$. Restrict to a function on $[0, L]$ and extend it to $[-L, L]$:

- the **even** extension kills all sine terms and gives the *half-range cosine series*

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right), \quad a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi}{L}x\right) dx; \quad (6)$$

- the **odd** extension kills all cosine terms and gives the half-range sine series.

COS uses the cosine version on a finite truncation interval $[a, b]$ (shift $x \mapsto x - a$, period $2(b - a)$). The reason for choosing cosine over the full complex series is taken up in §4.3.

3 The first Fourier pricing method: Carr–Madan (1998)

This is the precursor COS improves on; understanding it sharpens the contrast. Write $k = \ln K$, $x_T = \ln S_T$. The call is

$$C_T(k) = e^{-rT} \int_k^{\infty} (e^{x_T} - e^k) f(x_T | x_0) dx_T. \quad (7)$$

As $k \rightarrow -\infty$, $C_T(k) \rightarrow S_0$, so C_T is not square-integrable in k and cannot be Fourier-transformed directly. Carr–Madan *damp* it,

$$c_T(k) = e^{\alpha k} C_T(k), \quad \alpha > 0, \quad (8)$$

so $c_T \in L^1 \cap L^2$. Its Fourier transform $\psi_T(v) = \int_{-\infty}^{\infty} e^{ivk} c_T(k) dk$ can be written in closed form in terms of the characteristic function:

$$\psi_T(v) = \frac{e^{-rT} \varphi_X(v - (\alpha + 1)i)}{\alpha^2 + \alpha - v^2 + i(2\alpha + 1)v}, \quad (9)$$

and the price is recovered by an *inverse* transform, evaluated by FFT:

$$C_T(k) = \frac{e^{-\alpha k}}{2\pi} \int_{-\infty}^{\infty} e^{-ivk} \psi_T(v) dv. \quad (10)$$

Two structural weaknesses, both of which COS removes: (i) it still performs a numerical *integration* (the FFT computes a trapezoidal-type quadrature of an oscillatory integrand), and (ii) it needs a free damping parameter α , whose choice affects accuracy. COS needs neither.

4 Density recovery by the COS method

4.1 The cosine expansion and its coefficients

Let f be (essentially) supported on a finite interval $[a, b]$. Its half-range cosine series (6), written with the prime convention, is

$$f(x) = \sum_{k=0}^{\infty'} A_k \cos\left(k\pi \frac{x-a}{b-a}\right), \quad A_k = \frac{2}{b-a} \int_a^b f(x) \cos\left(k\pi \frac{x-a}{b-a}\right) dx, \quad (11)$$

where $\sum'$ halves the $k=0$ term (so $A_0/2$ is the mean of f over $[a, b]$).

4.2 The key identity: A_k from the characteristic function

This is the engine of the method. Start from the coefficient integral and write the cosine as the real part of a complex exponential:

$$\begin{aligned} A_k &= \frac{2}{b-a} \int_a^b f(x) \cos\left(k\pi \frac{x-a}{b-a}\right) dx = \frac{2}{b-a} \operatorname{Re} \left\{ \int_a^b f(x) \exp\left(ik\pi \frac{x-a}{b-a}\right) dx \right\} \\ &= \frac{2}{b-a} \operatorname{Re} \left\{ e^{-ik\pi \frac{a}{b-a}} \int_a^b f(x) \exp\left(i \frac{k\pi}{b-a} x\right) dx \right\}. \end{aligned} \quad (12)$$

Now *extend the integration range to $\mathbb{R}$* . Because f has negligible mass outside $[a, b]$,

$$\int_a^b f(x) e^{i \frac{k\pi}{b-a} x} dx \approx \int_{-\infty}^{\infty} f(x) e^{i \frac{k\pi}{b-a} x} dx = \varphi_X\left(\frac{k\pi}{b-a}\right), \quad (13)$$

the characteristic function (3) evaluated at $t = k\pi/(b-a)$. Substituting (13) into (12) gives the coefficient *without any integral*:

$$A_k \approx F_k := \frac{2}{b-a} \operatorname{Re} \left\{ \varphi_X\left(\frac{k\pi}{b-a}\right) \exp\left(-ik\pi \frac{a}{b-a}\right) \right\}. \quad (14)$$

The single approximation made so far is the range truncation in (13); everything else is exact. *This is precisely why COS needs only φ and not f .*

Worked check: the standard normal. For $X \sim \mathcal{N}(0, 1)$, $\varphi_X(t) = e^{-t^2/2}$. With $a = -b$ symmetric,

$$F_k = \frac{2}{b-a} \operatorname{Re} \left\{ e^{-\frac{1}{2} \left(\frac{k\pi}{b-a}\right)^2} e^{-ik\pi \frac{a}{b-a}} \right\},$$

matching the slide formula. A natural range is $a = \Phi^{-1}(10^{-8})$, $b = -a$ (clip the tails at probability 10^{-8}). With $K = 50$ terms the reconstruction is graphically indistinguishable from the true density, and at a fixed point the error falls *exponentially* in K until it saturates at the (fixed) range-truncation floor — the canonical COS convergence picture (§7).

4.3 Why a cosine series and not a full complex Fourier series

Three reasons, each worth stating explicitly because it is a favourite question:

1. **Real coefficients, real arithmetic.** The even extension of f on $[a, b]$ kills every sine term; the surviving coefficients A_k are real ((14) is a real part). A complex series would carry twice the coefficients and complex bookkeeping for no gain.
2. **Direct sampling from φ .** The cosine coefficient is exactly the real part of φ evaluated at the grid frequency $k\pi/(b-a)$ ((14)). The cosine basis is the one whose coefficients are “read off” from the characteristic function in one line.

3. **Payoff side closes in elementary functions.** The payoff coefficients V_k (§5.2) are integrals of (polynomial/exponential) $\times \cos$, which integrate to elementary closed forms χ_k, ψ_k . With a complex exponential basis the call payoff coefficients would not be as clean.

Equivalently: the cosine series is the discrete cosine transform analogue, and the whole method is a clever way to pair the cosine coefficients of f (got from φ) against the cosine coefficients of V (got in closed form) — see §5.

5 COS for European option pricing

5.1 Deriving the pricing formula

Insert the cosine expansion (11) of the (truncated) density into the price (1) and restrict the integral to $[a, b]$:

$$P(x_0, T) \approx e^{-rT} \int_a^b V(y, T) \sum_{k=0}^{\infty'} A_k(x_0) \cos\left(k\pi \frac{y-a}{b-a}\right) dy. \quad (15)$$

Interchange sum and integral (justified by uniform convergence on $[a, b]$ for smooth f) and define the **payoff cosine coefficients**

$$V_k := \frac{2}{b-a} \int_a^b V(y, T) \cos\left(k\pi \frac{y-a}{b-a}\right) dy. \quad (16)$$

Then

$$P(x_0, T) \approx e^{-rT} \frac{b-a}{2} \sum_{k=0}^{\infty'} A_k(x_0) V_k. \quad (17)$$

Finally replace A_k by its characteristic-function form (14) — note $\frac{b-a}{2} A_k = \operatorname{Re}\{\varphi_X(\frac{k\pi}{b-a}) e^{-ik\pi a/(b-a)}\}$ — and truncate the series at N terms:

$$P(x_0, T) \approx e^{-rT} \sum_{k=0}^{N-1'} \operatorname{Re}\left\{ \varphi_{X_T}\left(\frac{k\pi}{b-a}; x_0\right) \exp\left(-ik\pi \frac{a}{b-a}\right) \right\} V_k. \quad (18)$$

This is the European COS formula. The characteristic function enters in exactly one place: as the cosine coefficients of the density, $\operatorname{Re}\{\varphi e^{-ik\pi a/(b-a)}\}$. The dependence on the spot x_0 lives inside $\varphi_{X_T}(\cdot; x_0)$; for Lévy/BS models $\varphi_{X_T}(u; x_0) = \varphi_{\text{Lévy}}(u) e^{iu x_0}$, so a single set of $\{V_k\}$ prices the whole strike/spot grid — only the cheap $e^{iu x_0}$ factor changes. Cost is $O(N)$ per price; no integration, no FFT.

5.2 Closed form of the payoff coefficients V_k

Work in the log variable $y = \ln(S_T/K)$, so $S_T = K e^y$ and the strike sits at $y = 0$. Introduce the two building-block integrals on a sub-interval $[c, d] \subseteq [a, b]$, with $\omega_k := \frac{k\pi}{b-a}$:

$$\chi_k(c, d) := \int_c^d e^y \cos(\omega_k(y-a)) dy, \quad (19)$$

$$\psi_k(c, d) := \int_c^d \cos(\omega_k(y-a)) dy. \quad (20)$$

Derivation of χ_k . Write the cosine as a real part and integrate the complex exponential exactly:

$$\chi_k(c, d) = \operatorname{Re} \left\{ \int_c^d e^{(1+i\omega_k)y} e^{-i\omega_k a} dy \right\} = \operatorname{Re} \left\{ \frac{e^{-i\omega_k a}}{1+i\omega_k} \left[e^{(1+i\omega_k)y} \right]_c^d \right\}. \quad (21)$$

Multiply numerator and denominator by $(1-i\omega_k)$ (so the denominator becomes $1+\omega_k^2$) and take the real part of $e^y e^{i\omega_k(y-a)}(1-i\omega_k) = e^y [\cos \omega_k(y-a) + \omega_k \sin \omega_k(y-a)] + i(\dots)$. This gives the standard closed form

$$\chi_k(c, d) = \frac{1}{1+\omega_k^2} \left[\cos(\omega_k(d-a))e^d - \cos(\omega_k(c-a))e^c + \omega_k \sin(\omega_k(d-a))e^d - \omega_k \sin(\omega_k(c-a))e^c \right]. \quad (22)$$

Derivation of ψ_k . Elementary:

$$\psi_k(c, d) = \begin{cases} \frac{1}{\omega_k} [\sin(\omega_k(d-a)) - \sin(\omega_k(c-a))] = \frac{b-a}{k\pi} [\sin(\omega_k(d-a)) - \sin(\omega_k(c-a))], & k \neq 0, \\ d-c, & k = 0. \end{cases} \quad (23)$$

Call and put. For the call, $V = K(e^y - 1)^+$ has support $y \in [0, \infty)$, so on $[a, b]$ the relevant interval is $[0, b]$:

$$V_k^{\text{call}} = \frac{2}{b-a} K \int_0^b (e^y - 1) \cos(\omega_k(y-a)) dy = \frac{2}{b-a} K [\chi_k(0, b) - \psi_k(0, b)]. \quad (24)$$

For the put, $V = K(1 - e^y)^+$ has support $y \in (-\infty, 0]$, so the interval is $[a, 0]$:

$$V_k^{\text{put}} = \frac{2}{b-a} K \int_a^0 (1 - e^y) \cos(\omega_k(y-a)) dy = \frac{2}{b-a} K [\psi_k(a, 0) - \chi_k(a, 0)]. \quad (25)$$

These V_k are computed once. Put–call parity gives a fast cross-check and, in practice, the put is preferred when its density mass sits in a numerically gentler region; either way (18) returns the price.

5.3 Greeks at no extra cost

Because S_0 (equivalently x_0) enters (18) only analytically through $\varphi_{X_T}(\cdot; x_0)$, differentiation passes through the finite sum. With $\varphi_{X_T}(u; x_0) = \varphi(u) e^{iux}$, $x = \ln(S_0/K)$,

$$\Delta = \frac{\partial P}{\partial S_0} \approx e^{-rT} \sum_{k=0}^{N-1'} \operatorname{Re} \left\{ \varphi\left(\frac{k\pi}{b-a}\right) e^{ik\pi \frac{x-a}{b-a}} \cdot \frac{ik\pi}{b-a} \right\} \frac{V_k}{S_0}, \quad (26)$$

$$\text{Vega} = \frac{\partial P}{\partial u_0} \approx e^{-rT} \sum_{k=0}^{N-1'} \operatorname{Re} \left\{ \frac{\partial \varphi\left(\frac{k\pi}{b-a}; u_0\right)}{\partial u_0} e^{ik\pi \frac{x-a}{b-a}} \right\} V_k, \quad (27)$$

where u_0 is the relevant model parameter (e.g. the initial variance in Heston). No re-integration is needed — a structural advantage over quadrature- and FFT-based pricers.

6 The truncation range $[a, b]$ from cumulants

The only approximation in the density step was discarding the tails outside $[a, b]$ in (13). The range must therefore be wide enough to contain essentially all the probability mass of X_T , but

no wider (too wide wastes terms: a fixed accuracy needs $N \propto (b - a)$). Fang–Oosterlee fix it from the *cumulants* c_n of X_T :

$$\boxed{[a, b] = \left[c_1 - L\sqrt{c_2 + \sqrt{c_4}}, \quad c_1 + L\sqrt{c_2 + \sqrt{c_4}} \right], \quad L \in [8, 12] \text{ (typ. 10)}. \quad (28)}$$

Reading the terms:

- c_1 is the **mean** — it centres the interval where the mass is;
- c_2 is the **variance** — $\sqrt{c_2}$ is the natural width scale, so $L\sqrt{c_2}$ is an “ L -sigma” box;
- $\sqrt{c_4}$ (from the fourth cumulant, controlling tail/kurtosis) **fattens** the box for heavy-tailed models (jumps, VG); for Gaussian models $c_4 = 0$ and it drops out;
- L is the safety multiplier; larger L lowers the truncation floor but demands more terms.

Black–Scholes cumulants. For $X_T = \ln(S_T/K)$ under GBM, $\ln S_T$ is normal, so only c_1, c_2 are nonzero:

$$c_1 = \ln \frac{S_0}{K} + \left(r - \frac{1}{2}\sigma^2 \right) T, \quad c_2 = \sigma^2 T, \quad c_4 = 0, \quad (29)$$

giving the clean box $[a, b] = c_1 \pm L\sigma\sqrt{T}$.

6.1 The short-maturity failure (and why it bites)

This is exactly the kind of trap an examiner sets, and a bug I hit in Assignment 2. As $T \rightarrow 0$, $c_2 = \sigma^2 T \rightarrow 0$, so the half-width $L\sqrt{c_2 + \sqrt{c_4}} \rightarrow 0$ and $[a, b]$ *collapses* towards the single point c_1 . Consequences:

- the interval can become so narrow that it fails to contain the region where the payoff $\times$ density product has mass — in particular, in a Gaussian model with $c_4 = 0$ there is no $\sqrt{c_4}$ term to keep a floor, so the box shrinks unchecked;
- when pricing across a *grid* of spots/strikes, a range computed for one (S_0, K) may be centred by c_1 away from another point on the grid, so the truncated density is evaluated where the cosine reconstruction is meaningless — producing large, erratic pricing errors precisely at short maturities;
- because the range error is a *floor* on accuracy (§7), simply adding more cosine terms N does *not* fix it — the error plateaus.

Fixes: (i) increase L at short maturity; (ii) keep the $\sqrt{c_4}$ term and/or impose a minimum half-width floor so $b - a$ cannot collapse; (iii) make sure the range is computed to cover the whole spot/strike grid actually priced, not a single point. The diagnostic signature is: errors that *grow* as $T \rightarrow 0$ and that *do not* improve when N is increased — the tell-tale of a range, not a series, problem.

7 Error analysis and convergence

The COS price (18) carries three error sources, and the lecture’s central claim is that for smooth densities the total error converges *exponentially*.

1. **Range-truncation error** (from (13)): the discarded tail mass outside $[a, b]$. It is *fixed* once $[a, b]$ is chosen — it sets the accuracy *floor* at which the series convergence saturates. For densities with exponentially decaying tails it is itself exponentially small in the half-width.

2. **Series-truncation error** (cutting the sum at N): governed by the decay of the cosine coefficients A_k as $k \rightarrow \infty$. By the standard Fourier result, the smoother f , the faster A_k decay:
 - $f \in C^\infty$ with all derivatives integrable / f analytic in a strip $\Rightarrow$ **exponential** decay of $A_k \Rightarrow$ exponential convergence in N ;
 - f with only β integrable derivatives (a finite-smoothness density — e.g. near a jump, or VG with small ν) $\Rightarrow$ only **algebraic** decay $A_k = O(k^{-(\beta+1)}) \Rightarrow$ algebraic convergence. *Lack of smoothness is what kills the exponential rate.*
3. **Characteristic-function / model error**: any error in φ itself (e.g. a mis-evaluated Heston φ across the branch cut) feeds straight through.

The empirical picture (slide 24): at a fixed evaluation point the error drops like e^{-cN} and then *plateaus* at the range-truncation floor — raise the floor (wider $[a, b]$, larger L) and the plateau drops, but you then need more terms to reach it. Compared with Monte Carlo's $O(N^{-1/2})$ and Carr–Madan/FFT, COS reaches machine precision with N in the dozens to low hundreds (the Heston tables on slide 31: $N \sim 160$ for $\sim 10^{-10}$ error, versus thousands of FFT nodes for far worse error and longer CPU time).

8 COS versus Carr–Madan / FFT

	COS	Carr–Madan / FFT
Core operation	one $O(N)$ cosine sum, no integration	FFT quadrature, $O(N \log N)$
Free parameters	none (range from cumulants)	damping α to be tuned
Convergence (smooth f)	exponential in N	algebraic (quadrature-limited)
Terms for $\sim$ machine prec.	dozens–hundreds	thousands of nodes
Greeks	analytic, from the same sum	extra transforms / finite differences
Needs only φ ?	yes	yes

When COS wins: smooth (or piecewise-smooth) densities with known φ and exponentially decaying tails — i.e. most European pricing under BS, Heston, and the standard Lévy/AJD models. **When to be careful:** densities with genuine low regularity (slow algebraic decay of A_k) and very short maturities or fat tails, where $[a, b]$ must be set carefully (§6.1) and N raised.

9 Characteristic functions to keep at hand

From Fang–Oosterlee (2009); here ξ is the transform argument and t the horizon.

Model	Characteristic function
Black–Scholes	$\varphi(\xi, t) = \exp(i\xi\mu t - \frac{1}{2}\sigma^2\xi^2 t)$
NIG	$\varphi(\xi, t) = \exp[\delta t(\sqrt{\alpha^2 - \beta^2} - \sqrt{\alpha^2 - (\beta + i\xi)^2})]$
Kou	$\varphi(\xi, t) = \exp[\lambda t(\frac{p\eta_1}{\eta_1 - i\xi} + \frac{(1-p)\eta_2}{\eta_2 + i\xi} - 1)]$
Merton	$\varphi(\xi, t) = \exp[\lambda t(\exp(i\bar{\mu}\xi - \frac{1}{2}\bar{\sigma}^2\xi^2) - 1)]$
Variance-Gamma	$\varphi(\xi, t) = (1 - i\xi\theta\nu + \frac{1}{2}\sigma^2\nu\xi^2)^{-t/\nu}$
CGMY	$\varphi(\xi, t) = \exp(Ct\Gamma(-Y)[(M - i\xi)^Y - M^Y + (G + i\xi)^Y - G^Y])$

For BS the risk-neutral drift is $\mu = r - \frac{1}{2}\sigma^2$ in the log-price (martingale condition). Heston's φ (Lecture 7) plugs into the same formula (18); the Feller condition $2\kappa\bar{v} \geq \eta^2$ keeps the variance process from hitting zero and keeps φ well-behaved.

10 One-paragraph reconstruction (the thing to be able to say out loud)

The price is a discounted expectation $e^{-rT} \int V f$. I do not know f , but I know its Fourier transform, the characteristic function φ . So I truncate to a finite range $[a, b]$ chosen from the cumulants $(c_1 \pm L\sqrt{c_2 + \sqrt{c_4}})$, expand f there in a half-range *cosine* series whose coefficients are $A_k = \frac{2}{b-a} \operatorname{Re}\{\varphi(\frac{k\pi}{b-a})e^{-ik\pi a/(b-a)}\}$ — read straight off φ , no integral. I expand the payoff in the same cosine basis, getting coefficients V_k that are elementary closed forms (χ_k, ψ_k) . Plancherel-style, the integral of the product becomes $\frac{b-a}{2} \sum' A_k V_k$, i.e. $P \approx e^{-rT} \sum_{k=0}^{N-1} \operatorname{Re}\{\varphi(\frac{k\pi}{b-a})e^{-ik\pi a/(b-a)}\} V_k$. For smooth f the coefficients decay exponentially, so a few dozen terms reach machine precision; the only floor is the range truncation, which is why a too-narrow $[a, b]$ at short maturity ($c_2 = \sigma^2 T \rightarrow 0$) cannot be cured by adding terms.